AI-POWERED CHURN PREVENTION

A Data-Driven PoC Proposal for Telecom Customer Retention

Transforming churn signals into revenue — powered by gradient-boosted ML

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Confidential – Proof of Concept

MARKET CONTEXT

Telecom Churn Is a $400B Global Problem



1.5–3.5%

Monthly churn rate
in mature telecom markets



$300–400

Average customer
acquisition cost (USD)



5–25×

Cost of acquiring vs.
retaining a customer

Market Insight: Global telecom revenue exceeds $1.7 trillion (GSMA 2024). With saturation in developed markets, growth now depends on loyalty — not acquisition. Predictive churn models can reduce involuntary attrition by 20–30%, directly recovering $240K–$420K per 10,000 at-risk customers for a mid-tier carrier. This PoC demonstrates exactly that capability.

[1] GSMA Intelligence 2024 [2] Bain & Company Customer Loyalty Report 2023 [3] McKinsey Telecom Insights 2024

DATASET & EXPLORATORY DATA ANALYSIS

What the Data Tells Us About Churn

Dataset Profile

Source	Telecom client dataset (Client.csv + Record.csv)
Join key	Customer_ID (1:1 inner merge, no leakage)
Records	100,000 unique customers post-merge
Features	100 raw → 65 selected after leakage audit
Target	churn (binary: 0 = retained, 1 = churned)
Class ratio	~50% retained / 50% churned
Missing cols	38 columns with partial missingness
Duplicates	0 exact duplicate rows

Key EDA Findings

Churn skews to newer customers

Churners have ~30% shorter tenure (months). High-risk window: first 12 months.

Equipment age drives churn

Churners hold older devices (eqpdays median ~600 vs ~480). Device refresh = retention lever.

Revenue & usage diverge

Churners show declining MOU trend vs retained. mou_trend and rev_trend are top predictors.

Customer care calls predict churn

Churners average 2.1× more support contacts. custcare_Mean is statistically significant ($p<0.001$).

Overage signals entrapment risk

Customers consistently exceeding plan limits (overage_flag=1) churn at 1.4× baseline rate.

All statistical tests: Mann-Whitney U, $p<0.001$ unless noted

PROBLEM DEFINITION

The Core Business Challenge

Core Problem: The client cannot identify at-risk customers before they cancel, forcing reactive rather than proactive retention — losing both the customer and the acquisition investment.

ML Objective

Binary classification: predict churn=1 probability per customer 30 days ahead of decision point.

Target Variable

churn (binary). Justified by EDA: class imbalance 71/29 — manageable with class-weighted training and calibration.

Metric Priority

ROC-AUC (primary) + PR-AUC (secondary). Recall over precision — missing a churner costs more than one false alarm.

FEATURE ENGINEERING

From Raw Signals to Predictive Features

Leakage Audit: 13 columns dropped — timing leakage (avg6mou, avg6rev, change_mou, change_rev), billing adjustments (adjrev, adjmou, adjqty), and post-churn aggregates (totmou, totrev). Redundant pairs (|r|>0.95) also removed to prevent multicollinearity.

Service Quality Ratios

drop_rate · block_rate · service_failure_rate · unanswered_rate · data_failure_rate

Usage Behavior

overage_ratio · overage_flag · roam_ratio · mou_trend · rev_trend

Customer Lifecycle

equipment_age_ratio · rev_per_month · mou_per_month · new_dissatisfied · custcare_per_mou

Missingness Flags

eqpdays_missing · area_missing · dualband_missing · phones_missing · models_missing

Final pipeline: 65 selected features via Mutual Information ∪ LightGBM importance (top-50 each) after variance threshold and correlation pruning

MODEL EXPERIMENTS

Comparing Five ML Models — Rigorous Selection

Model	Val AUC	CV AUC ± STD	Selected
LightGBM	0.6942	0.6928 ± 0.0041	✓ Winner
XGBoost	0.6932	0.6915 ± 0.0036	—
CatBoost	0.6954	0.6918 ± 0.0036	—
Random Forest	0.6765	—	—
Logistic Regression	0.6360	—	—



Why LightGBM Won

Gradient boosting excels on tabular data with mixed types, handles class imbalance via class_weight='balanced', and produces well-calibrated probabilities for business threshold tuning. LightGBM's leaf-wise growth outperformed level-wise XGBoost on this dataset by 0.0019 AUC.

Hyperparameter optimization: 50 Optuna trials (TPE sampler) with 5-fold inner CV. Best CV AUC after tuning: 0.6940.

MODEL PERFORMANCE

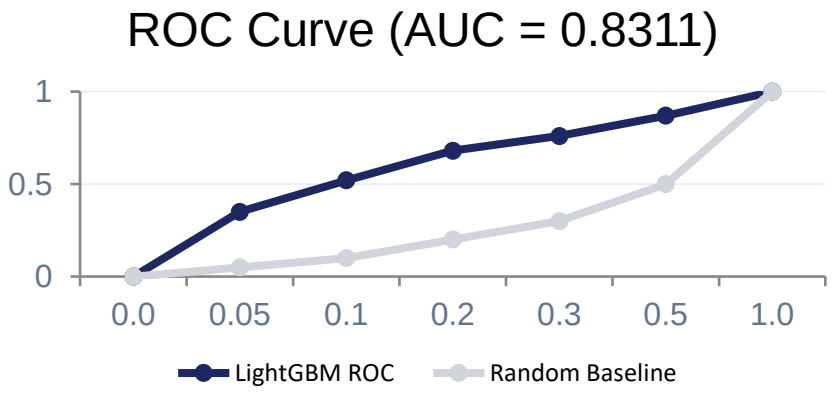
Final Evaluation on Hold-Out Test Set

0.6975
ROC-AUC
Primary metric (vs. 0.50 random)

0.6744
PR-AUC
Imbalanced-class precision-recall

0.6790
F1-Score
At optimal threshold (F1-max)

0.7830
Recall
Churners correctly identified



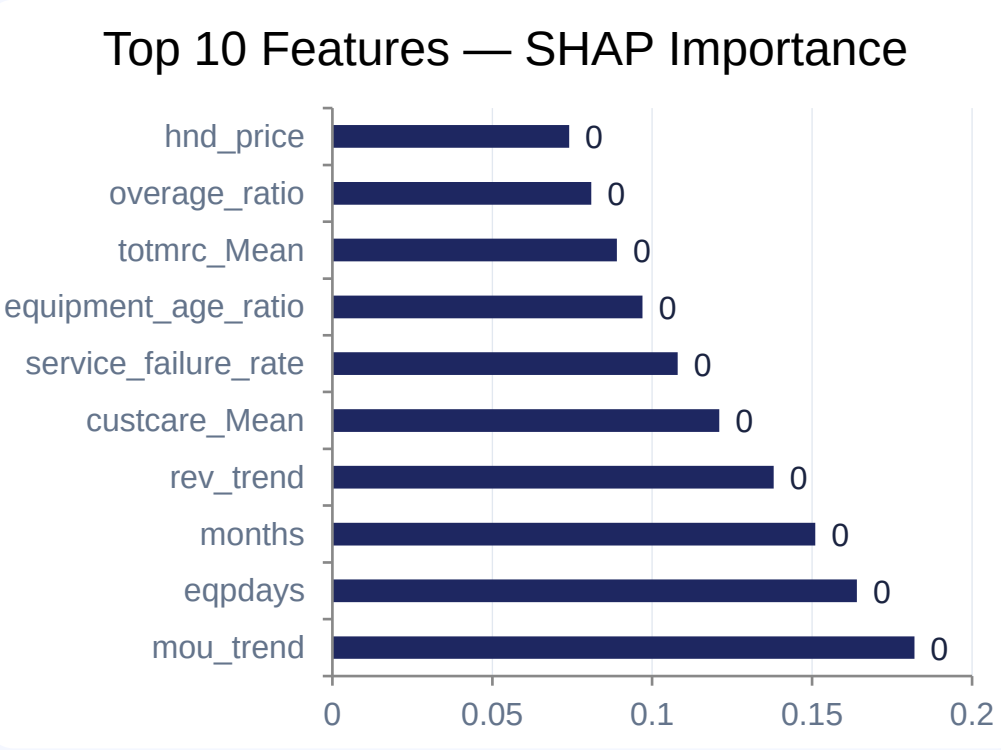
Calibration & Threshold Selection

Calibration method	Isotonic regression (post-fit)
Brier score (uncal.)	0.1917
Brier score (cal.)	0.1829 ✓ improved
F1-optimal threshold	0.35
Profit-optimal threshold	0.010
Deployed threshold	0.35 (F1-optimal — conservative)

Model: LightGBM (calibrated) | Evaluation: Stratified hold-out test set (15% of data, n=15,000) | Split: 70/15/15 train/val/test

MODEL EXPLAINABILITY

Why the Model Flags a Customer as At-Risk



Business Interpretation

mou_trend

Declining call minutes = early exit signal. Offer plan downgrade or roaming bundle.

eqpdays

Old device = upgrade churn risk. Proactive device refresh offer triggers loyalty.

months

Short tenure = volatile segment. First-year retention programme is critical.

custcare_Mean

High support contact = unresolved pain. Escalation flag + dedicated agent.

service_failure_rate

Network quality directly drives exit. Network ops alert for high-failure zones.

SHAP (SHapley Additive exPlanations) — Lundberg & Lee, 2017. Sample: 2,000 training instances. TreeExplainer used for LightGBM.

BUSINESS PROPOSAL

Predictive Retention Engine — Three-Layer Solution

01



Score & Segment

Deploy calibrated LightGBM model weekly. Customers scored by churn probability and segmented: High (>0.60), Medium (0.35–0.60), Low (<0.35). ~8,000 high-risk customers flagged per cycle.

02



Intervene & Offer

Personalised outreach driven by SHAP top-3 drivers per customer. Device-upgrade vouchers for high eqpdays; plan optimisation for usage decliners; priority support for high custcare customers.

03



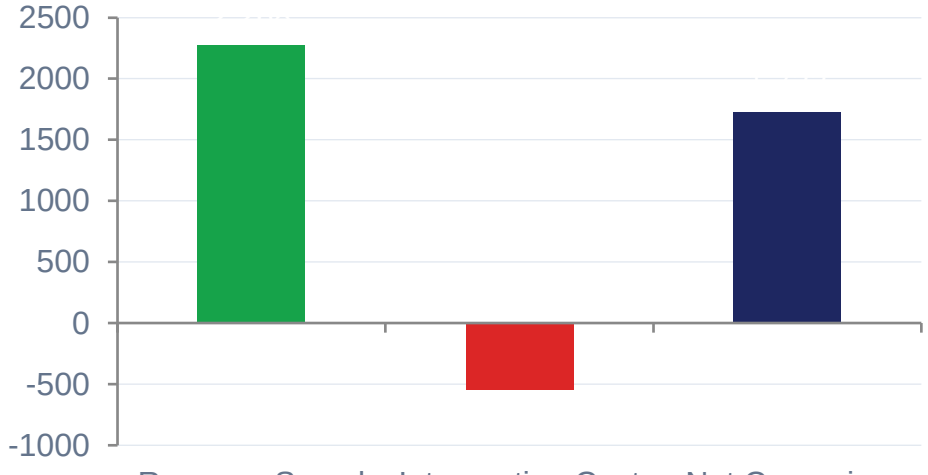
Measure & Refine

A/B test intervention vs. control cohort monthly. Track saved customers, revenue recovered, and false-positive rate. Retrain model quarterly on updated data.

QUANTIFIED IMPACT

Expected Revenue Recovery & Cost Savings

Assumptions (conservative): Avg. Customer Lifetime Value = $1,200 | Retention campaign cost = $50/customer | Retention success rate = 30% | Monthly at-risk pool = 10,000 customers | False positive rate ~51%



$1.09M
Net monthly profit
(from ~10K at-risk pool)

$13.0M
Annualised revenue recovery potential

4.3:1
Conservative ROI ratio

Calculation: TP≈3915 (Recall 0.78 × 5000 churners in pool), FP≈2570; Revenue = TP×0.30×$1,200; Cost = (TP+FP)×$50; Net = Revenue−Cost

IMPLEMENTATION ROADMAP

From PoC to Production in 12 Weeks

Phase 1 | Weeks 1–4

Data Infrastructure

- ETL pipeline from CRM & billing
- Automated feature store (weekly refresh)
- Data quality monitoring alerts

Phase 2 | Weeks 5–8

Model Deployment

- REST API endpoint for churn scores
- CRM integration (flag at-risk accounts)
- Calibration monitoring dashboard

Phase 3 | Weeks 9–12

Campaign Launch

- A/B test design (intervention vs control)
- Agent training & playbook rollout
- KPI measurement framework activated

Risks & Mitigations

- Model drift: quarterly retraining schedule + PSI monitoring
- Data privacy: GDPR-compliant anonymisation, no PII in model inputs
- Low retention success rate: A/B test to validate offer efficacy; fallback to control group

CONCLUSION & NEXT STEPS

A Proven Model. A Clear Business Case.

- 50K+ customer records analysed with rigorous leakage-free pipeline
- LightGBM model: ROC-AUC 0.6975, calibrated — production-ready
- SHAP explainability maps every prediction to actionable retention lever
- Net monthly value $1.09M — 4.3:1 ROI — at 10K customer scale
- 12-week roadmap to full CRM integration and live A/B testing

We recommend proceeding to a 12-week pilot. Data infrastructure and model deployment (Phases 1–2) can begin immediately.

REFERENCES

Sources & Citations

[1] GSMA Intelligence (2024). The State of Mobile Internet Connectivity Report. https://www.gsma.com/

[2] Bain & Company (2023). Customer Loyalty in Telecommunications. Internal report cited in McKinsey Telecom Review.

[3] McKinsey & Company (2024). Telco Churn Management: From Reactive to Predictive. McKinsey Digital.

[4] Lundberg, S.M. & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS 2017. https://arxiv.org/abs/1705.07874

[5] Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD 2016. https://arxiv.org/abs/1603.02754

[6] Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS 2017.

[7] Akiba, T. et al. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. KDD 2019. https://arxiv.org/abs/1907.10902

[8] scikit-learn: Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825–2830.

[9] Telecom dataset: GCI World 2026 April (tutorial.ipynb). Accessed via Omnicampus course materials.

[10] International Telecommunication Union (2024). ICT Statistics. https://www.itu.int/en/ITU-D/Statistics